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FoodEx2 annual maintenance 2015

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Abstract

FoodEx2 is a comprehensive food classification and description system released by EFSA in 2011 (revision 1) and replaced in April 2015 by FoodEx2 revision 2, as a consequence of a testing phase. Afterwards, the annual maintenance of the system was carried out in order to evaluate further comments and proposals provided to EFSA by users and stakeholders during the year 2015. This technical report describes the outcome of the maintenance process, including addition of new terms, disposal of existing terms and amendments to existing terms.

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Summary

A standardised food classification and description system is a key element to provide a common link to diverse information sources within different food safety domains. In order to match this need, in 2011 FoodEx2 (revision 1) was published by EFSA. After its first release, a major revision of the system was undertaken and in 2015 FoodEx2 revision 2 was published. The second revision achieved a good level of stability; however, innovation in the food market and accumulated experience by users lead to continuous need for updates in the system.

A maintenance technical group composed of EFSA staff with experience in different domains was set-up to evaluate and eventually implement proposals and requests provided by users and stakeholders during the year 2015.

Changes in the terminology and the structure of the system were applied: 271 new terms were added in different hierarchies and facets, the scope of some existing terms was better specified or expanded by changing the name or adding alias – included terms, 3 terms were dismissed because duplicated and the use of 9 terms was discouraged due to inconsistency with the logic of the system. The applicability (i.e. the usage) of some terms was extended to further hierarchies and/or facets and the position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found. Furthermore the structure and content of the FoodEx2 sections related to tea and herbal/fruit infusions were refined in order to align them with the Regulation (EC) No 396/2005 and with the THIE Compendium of Guidelines for Tea (*Camellia sinensis*).

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1. Introduction

1.1. Background and Terms of Reference as provided by the requestor

In 2011, EFSA released a comprehensive Food Classification and Description System (named FoodEx2 revision1) (EFSA, 2011a,b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), the users should provide to EFSA proposals for implementation and maintenance in order to refine the system and an annual EFSA maintenance working group should be set up in order to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system lead to the need of keeping the classification updated.

By using the system, Member States and other stakeholders, including internal users in EFSA, provided constructive comments and requests for addition of missing terms. These suggestions were taken into account for the maintenance purposes. A technical group composed of EFSA staff with experience in different domains carried out the yearly maintenance process evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the maintenance process in order to keep track of the changes in the system.

2. Data and Methodologies

2.1. Data

The FoodEx2 revision2 catalogue published in April 2015 was used as starting point. The suggestions provided during the year 2015 by the users and stakeholders of the FoodEx2 system were collected in a working document; this document was used to perform the maintenance.

2.2. Methodologies

Each request for addition or change in the structure was considered and, on the basis of expert judgment, the maintenance group decided whether a change was needed or whether the request was already covered by the available terms and facet descriptors. All the changes were made by following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1. Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Browser, a user friendly tool developed in Java by staff of EFSA (DATA Unit) to help creating FoodEx2 codes and to facilitate the management of the catalogue¹.

2.2.2. Maintenance process

In order to carry out the yearly maintenance, the maintenance group performed the listed steps:

- Evaluating comments and proposals;
- Deciding on the implementation of the proposals; different options were available: adding new terms, extending or better specifying the scope of existing terms or identifying the correct way of implementing the suggestion using already existing terms and rules;
- Checking for inconsistencies and ambiguities;
- Updating the FoodEx2 catalogue;
- Summarizing the changes for the present technical report.

¹ Available online at <http://www.efsa.europa.eu/en/datex/datexfoodclass.htm>

3. Maintenance output

3.1. Addition of new terms

The terms added in the 2015 annual maintenance were in total 271. The Reporting hierarchy gained 82 new terms of which 47 in the Food section and 35 in the Feed section; the new terms of the Food section are reported in Table 1 organized by the aggregated groups of the Exposure hierarchy, while the ones of the Feed section are presented in Table 2 organized by the aggregated groups of the Feed hierarchy. Ninety-two new terms, in particular organisms relevant for vector-borne diseases, were added only in the Master hierarchy. These terms are not listed in the present report because not relevant as food or as target for zoonoses. The remaining 97 terms were added as facet descriptors: 18 terms as part-nature descriptors, 2 terms as process descriptors, 5 terms as qualitative-info descriptors, 1 term as surrounding-medium descriptor, 1 term as purpose-of-raising descriptor, 69 terms as source descriptors and 1 term as source-commodities descriptor (Table 3).

Table 1: New food terms added in the 2015 maintenance. The terms are organized by the aggregated groups of the Exposure hierarchy; the relevant aggregated groups of the Exposure hierarchy are also included in the table (grey italic entries)

New name		Corex ^(b)	State ^(b)	Motivation and reason
Code	Name			
<i>A000J</i>	<i>Grains and grain-based products</i>	<i>H</i>	<i>g</i>	
A16FX	Fried bread dough	C	s	Fried dough products are commonly found in many European countries
A16FH	Phyllo dough	E	s	Typical dough in Middle Eastern and Balkan cuisines with important difference respect to the generic puff pastry dough
A16FJ	Sponge biscuits	E	s	Egg-based biscuits characterized by their low density
A16FV	Flat bread-based pastry	C	c	Term describing typical traditional foods e.g. karelian pastry, flaouna and similar
A16FY	Fried dough sweet	C	c	Commonly found in many European countries
A16EQ	Puddings based on cereal products or bread	C	c	Term describing puddings based on cereal products or bread, previously coded with more generic terms because a specific term was missing.
A16GM	Mixed roasted flours for porridge-like food	C	s	Term to describe e.g. traditional Estonian and Finnish finely milled roasted flour mixture, not well covered by existing terms
<i>A00ZR</i>	<i>Starchy roots or tubers and products thereof, sugar plants</i>	<i>H</i>	<i>g</i>	
A16FL	Chinese water chestnut	E	r	Raw commodity obtained from the plant species <i>Eleocharis dulcis</i> . The plant is mainly grown for its edible corms.
A16FM	Canned starchy root products	E	d	Term to describe starchy roots and tubers canned
<i>A011X</i>	<i>Legumes, nuts, oilseeds and spices</i>	<i>H</i>	<i>g</i>	
A16FZ	Chia seeds	E	r	Raw commodity obtained from the plant species <i>Salvia hispanica</i>
<i>A01BS</i>	<i>Fruit and fruit products</i>	<i>H</i>	<i>g</i>	
A16FC	Fruit jelly	C	s	Term describing fruit jellies according to Council Directive 2001/113/EC ²
A16GL	Fruit dessert thickened	C	s	Term describing viscous fruit dishes (e.g. kissel, sugo d'uva and rote gruetze) commonly found in many European countries

² Council Directive 2001/113/EC of 20 December 2001 relating to fruit jams, jellies and marmalades and sweetened chestnut purée intended for human consumption. OJ L 10, 12.1.2002, p. 67–72.

New name				
Code	Name	Corex ^(b)	State ^(b)	Motivation and reason
<i>A01QR</i>	<i>Meat and meat products</i>	<i>H</i>	<i>r</i>	
A16EL	Ciccioli and similar	C	d	Traditional food item based on animal fat tissue commonly found in many European countries
<i>A026T</i>	<i>Fish, seafood, amphibians, reptiles and invertebrates</i>	<i>H</i>	<i>g</i>	
A16GD	Tench	C	r	Raw commodity obtained from the fish species <i>Tinca tinca</i>
A16GE	River lamprey	C	r	Raw commodity obtained from the fish species <i>Lampetra fluviatilis</i>
A16FS	Herring, 8eonat	E	r	Term identifying organisms of the species <i>Clupea harengus</i> living in the Baltic sea. This herrings are characterized by smaller size and less fat than true Atlantic herrings
A16FR	Seafood offal	M	r	Term describing offal from seafood
A16FA	Fermented fish	C	d	Traditional preparation of fish found in different European countries
A16EM	Smoked mackerel	C	d	Traditional food dish commonly found in Northern Europe
<i>A02LR</i>	<i>Milk and dairy products</i>	<i>H</i>	<i>g</i>	
A16GH	Strained yoghurt	C	d	Term describing strained yoghurt such as Greek yoghurt
A16FE	Cheese powder	C	d	Term describing any type of grated cheese
<i>A039K</i>	<i>Fruit and vegetable juices and nectars (including concentrates)</i>	<i>H</i>	<i>g</i>	
A16MQ	Juice concentrate, apple	C	d	Term describing apple juice concentrates, widely available on European market
A16MR	Juice concentrate, pineapple	C	d	Term describing pineapple juice concentrates, widely available on European market
A16MS	Juice concentrate, pomegranate	C	d	Term describing pomegranate juice concentrates, widely available on European market
<i>A03LZ</i>	<i>Alcoholic beverages</i>	<i>H</i>	<i>g</i>	
A16GG	Root beer	C	s	Traditional beer-like beverage commonly found in some European countries
<i>A03GG</i>	<i>Coffee, cocoa, tea and infusions</i>	<i>H</i>	<i>g</i>	
A16GP ^(a)	Tea extract (liquid)	E	d	Term describing tea extracts in liquid form
A16GX ^(a)	Herbal infusion extracts (RPC derivatives)	C	d	Term describing herbal extracts used for making infusions
A16GY ^(a)	Herbal infusion extracts (powder)	E	d	Term describing herbal extracts in powder form used for making infusions
A16GZ ^(a)	Herbal infusion extracts (liquid)	E	d	Term describing herbal extracts in liquid form used for making infusions
A16GQ	Peppermint infusion	C	s	Term describing peppermint infusion commonly consumed in some European countries
A16GR	Fennel infusion	C	s	Term describing fennel infusion commonly consumed in some European countries
<i>A03PV</i>	<i>Food products for young population</i>	<i>H</i>	<i>g</i>	
A16GS	Herbal infusions specific for infants and young children, dry	C	s	Term describing herbal preparations in dried form used for making herbal infusions for infants and young children.
<i>A03VA</i>	<i>Composite dishes</i>	<i>H</i>	<i>c</i>	
A16ER	Offal based dishes	E	c	Term describing offal based dishes
A16FF	Bread- (or similar products) based	C	c	Term describing bread based composite dishes commonly found in many European countries (e.g.

New name		Corex ^(b)	State ^(b)	Motivation and reason
Code	Name			
	dishes			drained knoedel)
A16GB	Mixed vegetables	M	c	Term describing mixed vegetable preparations such as pre-prepared frozen cut vegetables to make soup
A16GF	Rice, fish and vegetable based dishes	E	s	Term describing rice, fish and vegetable based dishes such as sushi, when it includes rice
<i>A042N</i>	<i>Seasoning, sauces and condiments</i>	H	g	
A16GK	Savoury sauce dry preparation	C	s	Term describing dry preparations used to make savoury sauce
A16BS	Alcoholic sauce	C	s	Term describing sauces based on alcoholic ingredients
A16EY	Sweet and sour sauce	C	s	Term describing the typical Chinese sweet and sour sauce and similar products
A16BR	Meat sauce	C	s	Term describing meat based sauces
<i>A046L</i>	<i>Major isolated ingredients, additives, flavours, baking and processing aids</i>	H	g	
A16EN	Collagen	C	d	Term describing collagen, often found as supplement or fortification agent
A16EP	Hyaluronic acid	C	d	Term describing hyaluronic acid, often found as supplement or fortification agent
A16GN	Gluten	C	d	Term describing gluten, used as ingredient in some products
A169T	Raising agents	C	d	Term describing raising agents as food additive
A16GC	Preparations for dessert	C	s	Term describing home preparation aids for making desserts
A16FP	Probiotic cultures	C	r	Term describing probiotic microbial cultures, often found as supplement or fortification agent
<i>A0F0S</i>	<i>Other ingredients</i>	<i>H</i>	<i>r</i>	
A16GT	Taurine	C	d	Term describing taurine, often found as supplement or fortification agent

(a): See section 3.3 for further details on amendments on tea and herbal infusion sections

(b): Corex and State attributes of the FoodEx2 terms describing the level of detail and the type (raw commodity, derivative, composite etc.), respectively

Table 2: New feed terms added in the 2015 maintenance. The terms are organized by the aggregated groups of the Feed hierarchy; the relevant aggregated groups of the Feed hierarchy are also included in the table (grey italic entries)

New name		Corex ^(a)	State ^(a)
Code	Name		
<i>A0BT0</i>	<i>COMPOUND FEED</i>	<i>P</i>	<i>c</i>
<i>A0BT1</i>	<i>Complete feed</i>		
A16AC	Piglet Starter I (pre-starter) (weaning diets) / Complete feed	E	c
A16AD	Piglet Starter II (weaning diets) / Complete feed	E	c
A16AE	Pig Grower / Complete feed	E	c
A16AF	Pig Finisher / Complete feed	E	c
A16AG	Sows / Complete feed	C	c
A16AH	Gestating sows / Complete feed	E	c
A16BF	Sheep / Complete feed	C	c
A16BG	Goat / Complete feed	C	c
A16AJ	Broiler Starter / Complete feed	E	c
A16AK	Broiler Grower / Complete feed	E	c
A16AL	Broiler Finisher / Complete feed	E	c
A16AM	Layer Phase I / Complete feed	E	c

New name			
Code	Name	Corex ^(a)	State ^(a)
A16AN	Layer Phase II / Complete feed	E	c
A16BE	Chicken reared for laying / Complete feed	C	c
A16AP	Turkey Starter / Complete feed	E	c
A16AQ	Turkey Grower / Complete feed	E	c
A16AR	Turkey Finisher / Complete feed	E	c
A16AS	Turkey Breeder/ Complete feed	E	c
A16AT	Rabbit, breeder/ Complete feed	E	c
A16AV	Rabbit, grower-finisher/ Complete feed	E	c
A16AX	Salmon / Complete feed	E	c
A16AZ	Trout / Complete feed	E	c
A16AY	Carp / Complete feed	E	c
A16BA	Dog food (dry)/ Complete feed	E	c
A16BB	Dog food (wet)/ Complete feed	E	c
A16BC	Cat food (dry)/ Complete feed	E	c
A16BD	Cat food (wet)/ Complete feed	E	c
A0BV1	<i>Complementary feed (incomplete diet)</i>		
A16BH	Calves (pre-ruminant)-Mineral / Complementary feed	C	c
A16BJ	Fattening cattle-Mineral / Complementary feed	C	c
A16BK	Dairy cows-Mineral / Complementary feed	C	c
A16BL	Sheep / Complementary feed	C	c
A16BN	Sheep-Mineral / Complementary feed	C	c
A16BM	Goat / Complementary feed	C	c
A16BP	Goat-Mineral / Complementary feed	C	c
A16BQ	Horses –Mineral/ Complementary feed	C	c

(a): Corex and State attributes of the FoodEx2 terms describing the level of detail and the type (raw commodity, derivative, composite etc.), respectively

Table 3: New terms added as facet descriptors in the 2015 maintenance. The relevant facet for each of the new entries is reported in the column 'Facet'

Facet	New name		Corex ^(b)	State ^(b)	Motivation
	Code	Name			
Process	A16ES	Cold smoking	E	f	Terms describing two different types of smoking process
Process	A16ET	Hot smoking	E	f	
Purpose-of-raising	A16CZ	Fur animal	F	f	Term describing the practice of raising certain animals for their fur
Source	A16FK	Chinese water chestnut (live plants)	E	n	Term describing the plant species <i>Eleocharis dulcis</i> mainly grown for its edible corms
Source	A16GA	Chia (live plants)	E	n	Term describing the plant species <i>Salvia hispanica</i> mainly grown for its edible seeds
Source	A16BX	Antelopes (live animals)	E	n	Terms describing live animals mainly relevant in the zoonosis domain
Source	A16BY	Oryx antelopes (live animals)	E	n	
Source	A16CF	Barbary sheep (live animals)	E	n	
Source	A16DL	Mountain goat (live animals)	E	n	
Source	A16HN	Mountain hare (live animals)	E	n	
Source	A16HL	Vicugna (live animals)	E	n	
Source	A16HM	Sika deer (live animals)	E	n	
Source	A16CP	Dolphins (live animals)	E	n	
Source	A16DZ	Seals (live animals)	E	n	
Source	A16DY	Sea lions (live animals)	E	n	
Source	A16DM	Muskrat (live animals)	E	n	
Source	A16EE	Voles (live animals)	E	n	
Source	A16EB	Squirrels (live animals)	E	n	
Source	A16CJ	Beavers (live animals)	E	n	
Source	A16CN	Chinchillas (live animals)	E	n	
Source	A16DA	Gerbils (live animals)	E	n	
Source	A16DN	Octodons (live animals)	E	n	
Source	A16CA	Mustelidae (live animals)	E	n	
Source	A16CB	Badgers (live animals)	E	n	
Source	A16CD	Martens (live animals)	E	n	
Source	A16CE	Weasel (live animals)	E	n	
Source	A16CC	Ferret (live animals)	E	n	
Source	A16CL	Felidae (live animals)	E	n	
Source	A16CM	Cheetahs (live animals)	E	n	
Source	A16EG	Wild cat (live animals)	E	n	
Source	A16DF	Jaguar (live animals)	E	n	
Source	A16DH	Leopard (live animals)	E	n	
Source	A16DJ	Lion (live animals)	E	n	
Source	A16EC	Tiger (live animals)	E	n	
Source	A16DK	Lynx (live animals)	E	n	
Source	A16CX	Canidae (live animals)	E	n	
Source	A16EH	Wolf (live animals)	E	n	
Source	A16DE	Jackals (live animals)	E	n	
Source	A16CY	Foxes (live animals)	E	n	
Source	A16DR	Raccoon dog (live animals)	E	n	
Source	A16BZ	African Wild Dog (live animals)	E	n	
Source	A16CH	Bears (live animals)	E	n	
Source	A16EJ	Polar bear (live animals)	E	n	
Source	A16DS	Raccoons (live animals)	E	n	
Source	A16DG	Tree-kangaroo (live animals)	E	n	
Source	A16DV	Rufous rat-kangaroo (live animals)	E	n	
Source	A16CT	Elephants (live animals)	E	n	
Source	A16DB	Giraffe (live animals)	E	n	

Facet	New name		Corex ^(b)	State ^(b)	Motivation
	Code	Name			
Source	A16DT	Rhinoceros (live animals)	E	n	
Source	A16EA	Shrew (live animals)	E	n	
Source	A16CG	Bats (live animals)	E	n	
Source	A16DD	Hedgehog (live animals)	E	n	
Source	A16CQ	Accipitriformes (live animals)	E	n	
Source	A16CR	Eagles (live animals)	E	n	
Source	A16EK	White-tailed eagle (live animals)	E	n	
Source	A16CS	Cathartiformes (live animals)	E	n	
Source	A169V	Icterids (live animals)	E	n	
Source	A169X	Red-winged blackbird (live animals)	E	n	
Source	A16CK	Canary (live animals)	E	n	
Source	A16CV	Finches (live animals)	E	n	
Source	A16ED	Toucans (live animals)	E	n	
Source	A16EF	Swans (live animals)	E	n	
Source	A16DP	Parrots (live animals)	E	n	
Source	A16DQ	Penguins (live animals)	E	n	
Source	A16DC	Ground hornbill (live animals)	E	n	
Source	A16FT	Baltic herring (live animals)	E	n	
Source	A16DX	Salamanders (live animals)	E	n	
Source	A16BT	Schizosaccharomycetaceae (live organisms)	E	n	Term describing live organisms belonging to the family Schizosaccharomycetaceae
Source	A16BV	Schizosaccharomyces pombe (live organisms)	E	n	Term describing live organisms belonging to the species <i>Schizosaccharomyces pombe</i>
Source	A169Y	Sordariales (live organisms)	E	n	Term describing live organisms belonging to the order Sordariales
Source	A169Z	Sordaria brevicollis (live organisms)	E	n	Term describing live organisms belonging to the species <i>Sordaria brevicollis</i>
Source	A169M	Scenedesmus obliquus (live organisms)	E	n	Term describing live organisms belonging to the species <i>Scenedesmus obliquus</i>
Part-nature	A16FD	Fruit jellies (as part-nature)	C	f	Term describing the nature of fruit jellies
Part-nature	A16FB	Fermented fish (as part-nature)	C	f	Term describing the nature of fermented fishes
Part-nature	A16GJ	Bread- (or similar products) based dishes (as part-nature)	C	f	Term describing the nature of bread- (or similar products) based dishes
Part-nature	A16FG	Neither water- nor dairy-based spoonable dessert (as part-nature)	C	f	Term describing the nature of neither water- nor dairy- based spoonable desserts
Part-nature	A16EV	Alcoholic sauce (as part-nature)	C	f	Term describing the nature of alcoholic sauces
Part-nature	A16EX	Meat sauce (as part-nature)	C	f	Term describing the nature of meat sauces
Part-nature	A16FN	Ear (as part-nature)	E	f	Terms describing parts of animals
Part-nature	A16AA	Salivary glands (as part-nature)	E	f	Terms describing the nature of non-food animal-related matrices
Part-nature	A16HA	Fetus / stillbirth / 12 neonates (as part-nature)	F	f	
Part-nature	A16HC	Nose / nasal swab (as part-nature)	F	f	

Facet	New name		Corex ^(b)	State ^(b)	Motivation
	Code	Name			
Part-nature	A16HB	Esophagus / esophageal swab (as part-nature)	F	f	
Part-nature	A16HD	Placenta (as part-nature)	F	f	
Part-nature	A16HE	Rectum-anal swab (as part-nature)	F	f	
Part-nature	A16HF	Saliva / saliva swab (as part-nature)	F	f	
Part-nature	A16HG	Secreta (as part-nature)	F	f	
Part-nature	A16HH	Semen (as part-nature)	F	f	
Part-nature	A16HJ	Tonsil (as part-nature)	F	g	
Part-nature	A16HK	Vagina / vaginal swab (as part-nature)	F	g	
Surrounding-medium	A169N	In own juice	F	f	Term describing a surrounding medium
Qualitative-info	A16FQ	With probiotics and/or prebiotics	F	f	Term describing food products with probiotics and/or prebiotics
Qualitative-info	A16EZ	Savoury	F	f	Term describing the savoury taste
Qualitative-info	A169P	Blue	F	f	Term describing blue colour
Qualitative-info	A169Q	Semi-soft	F	f	Term describing semi-soft texture
Qualitative-info	A169R	Mixed origin	F	f	Term describing food products from multiple origins
Source-commodities	A16GV ^(a)	Camellia sinensis (tea) fresh leaves	H	r	Term describing fresh tea leaves as raw primary commodity

(a): See section 3.3 for further details on amendments on tea and herbal infusion sections

(b): Corex and State attributes of the FoodEx2 terms describing the level of detail and the type (raw commodity, derivative, composite etc.), respectively

3.2. Changes in existing terms

Taking into account the suggestions from the users and keeping the consistency of the system, amendments to existing terms were also applied.

Some new terms requested for addition were only added as common names/aliases of existing terms because they were already covered by the FoodEx2 catalogue content. These terms are listed in Table 4.

Table 4: Names that in the 2015 maintenance were added as alias (common names or included terms) of existing terms. The names of existing terms are shown in grey

Existing term		New alias – included term
Code	Name	
A0EVN	Carotenoids	Lutein
A048Q	Preparations for raising and flavouring home-made bakery	Baking powder
A001F	Rice grain, long-grain	Jasmine rice
A036V	Rape seed oil, edible	Canola oil
A00CC	Pastry based on laminated dough	Danish pastry
A01BA	Mushy peas	Peas pudding
A024J	Fresh spiced sausages in casing	Andouillette
A042Y	Seasoning mixes	Miso
A0BYR	Wafers	Raffaello
A04KK ^(a)	Teas leaves, dry and/or fermented, and similar	Tea: dried leaves, stalks and flowers, whether fermented or otherwise treated (Camellia sinensis)

Existing term		
Code	Name	New alias – included term
A0CSP	Steinbock (live animals)	Bouquetin Capricorn

(a): See section 3.3 for further details on amendments on tea and herbal infusion sections

In other cases, for some existing terms the name was changed to better specify the scope of the term or to expand its meaning; however, in some cases the old names were kept as alias – included terms. Table 5 shows the terms in which the name has been changed. During this phase, editorial changes, such as singular/plural or correction of misspelling, were also applied to several terms of FoodEx2. These minor editorials are not recorded in the present report but they were implemented, therefore they are included in the FoodEx database and appear in the Appendix file 'FoodEx2_may2016.xlsx' containing the complete content of the FoodEx2 system.

Table 5: New names assigned to existing terms to better specify the scope of the term or to expand its meaning. The old name is shown in grey

Term		
Code	New name	Old name
A008X	Short pastry dough and similar	Short pastry doughs (pate brisee)
A00CC	Pastry based on laminated dough	Puff pastry
A00HZ	Peppers and similar-	Sweet peppers and similar-
A00KX	Lettuces (generic)	Lettuces
A00XQ	Lemon myrtle (fresh herb)	Lemon myrtle
A019X	Cassia bark (generic)	Cassia bark
A01FE	Bilberries (generic)	Bilberries
A01QV	Bovine fresh meat	Bovine muscle
A01RG	Pig fresh meat	Pig muscle
A01RH	Sheep fresh meat	Sheep muscle
A01RL	Goat fresh meat	Goat muscle
A01RM	Equine fresh meat	Equine muscle
A02MH	Milk from other milk producer animals	Other milk producer animals
A03GH ^(a)	Ingredients for coffee, cocoa, tea, and herbal infusions	Coffee, cocoa, tea and herbal ingredients
A03HQ ^(a)	Tea leaves derivatives and tea ingredients	Tea leaves and tea ingredients
A03HY ^(a)	Tea extract (powder)	Instant tea powder
A03HZ ^(a)	Tea extract (powder) decaffeinated	Instant tea powder decaffeinated
A03JB ^(a)	Flowers used for herbal infusions	Herbal infusion materials from flowers
A03JS ^(a)	Roots used for herbal infusions	Herbal infusion materials from roots
A03JZ ^(a)	Hot drinks and similar (coffee, cocoa, tea and herbal infusions)	Hot drinks and similar (coffee, cocoa, tea and herbal drinks)
A03LC ^(a)	Fermented tea infusion	Tea infusion (black, white)
A03LL	Hibiscus infusion ^(b)	Roselle infusion
A03RM ^(a)	Herbal infusions (beverages) specific for infants and young children, liquid	Teas and infusions specific for infants and young children
A04KK ^(a)	Teas leaves, dry and/or fermented, and similar	Teas and similar-
A0BYA ^(a)	Derivatives of coffee, cocoa, tea, herbal infusion materials and similar RPCs	Derivatives of coffee, cocoa, tea and similar RPCs
A0C6A	Coffee ingredients (RPC derivatives)	Coffee RPC derivatives
A06DS	Pastry based on laminated dough (as part-nature)	Puff pastry (as part-nature)
A041Y	Cereal products and grains based soup	Grains soup
A0EYN	Preserved/processed fat tissues	Preserved fat tissues
A06AQ	Preserved/processed fat tissues (as part-nature)	Preserved fat tissues (as part-nature)
A043C	Indian spice mixes and similar (other than curry powder)	Garam masala
A03LD	Non-fermented tea, infusion	Green tea, infusion
A03HT	Non-fermented tea leaves (green or white tea)	Green tea leaves
A00EJ	Muesli and similar mixed breakfast cereals	Muesli and similar
A00EL	Mixed breakfast cereals	Muesli mixed
A06DZ	Muesli and similar mixed breakfast cereals (as part-nature)	Muesli (as part-nature)

Term		
Code	New name	Old name
A0FOX	Fermented/pickled/marinated vegetables (as part-nature)	Fermented/pickled/marinated vegetables

(a): See section 3.3 for further details on amendments on tea and herbal infusion sections

(b): Alias Karkadé and Roselle

The scope of some terms (i.e. 'Scopenotes') was amended to better explain the meaning of the code and to clarify which food items are covered by the code and which are not. However, broader improvements in the scope note description is planned for a separate future activity.

3.3. Amendments on tea and herbal infusion sections

The structure and content of the FoodEx2 sections related to tea and herbal/fruit infusions were refined on the basis of information provided to EFSA by the association Tea & Herbal Infusions Europe (THIE). The aim of the maintenance was to align the area of tea and herbal/fruit infusion ingredients and beverages with the Regulation (EC) No 396/2005³ and with the Compendium of Guidelines for Tea (*Camellia Sinensis*) (THIE, 2014), considering the practices of the tea industry and its market. So far, as explained in section 5.4.1 of the food classification and description system FoodEx2 (revision 2), in the case of agricultural products usually marketed as dry (e.g. spices and infusion materials, including tea) and normally not available in the fresh form in the European market, it was decided to consider as raw primary commodity the dried form instead of the fresh one. This rule is still valid for spices and herbal infusion materials, but for the tea (*Camellia sinensis*) it was preferred to move to the 'standard' approach (fresh raw commodity → derivatives) because tea is not only dried but in most cases it is also fermented. As already described in section 3.1, the fresh leaves of *Camellia sinensis* have been introduced as raw primary commodity (A16GV); from this commodity tea ingredients are obtained (mainly by fermentation and/or dehydration). Furthermore, a clearer distinction between fermented and unfermented teas and between tea and herbal extracts in liquid and powder form has been introduced. As a consequence, the sections on tea leaves and herbal materials used for infusions and their related beverages have been re-organized. A snapshot of FoodEx2 browser related to ingredients for tea and herbal infusions and a second one related to tea and herbal infusion beverages are shown in Figure 1 and 2, respectively.

³ Regulation (EC) No 396/2005 of the European Parliament and of the Council of 23 February 2005 on maximum residue levels of pesticides in or on food and feed of plant and animal origin and amending Council Directive 91/414/EEC Text with EEA relevance. OJ L 70, 16.3.2005, p. 1–16.

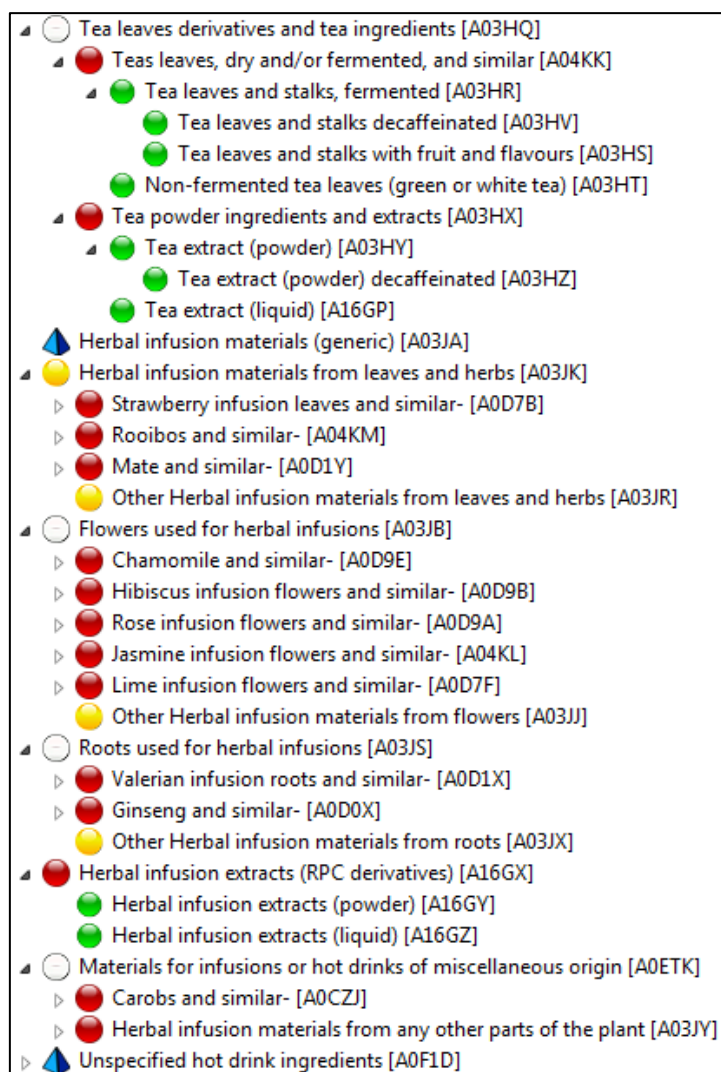


Figure 1: Snapshot of the FoodEx2 Browser (Exposure hierarchy) showing ingredients for tea and herbal infusions.

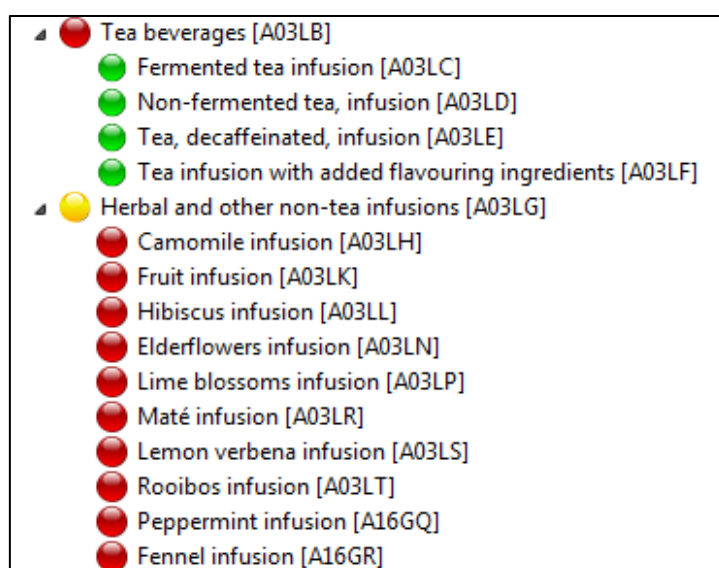


Figure 2: Snapshot of the FoodEx2 Browser (Exposure hierarchy) showing tea and herbal infusion beverages.

3.4. Terms that were dismissed or candidate for future dismissing

Three terms were dismissed while refining the tea and herbal infusion sections, because found to be double (see Table 6): the term A0C6C 'Tea ingredients (RPC derivatives)' had the same scope of A03HQ 'Tea leaves derivatives and tea ingredients', A0ETA 'Flowers used for infusions' the same as A03JB 'Flowers used for herbal infusions' and A0ETH 'Roots used for infusions' was the same as A03JS 'Roots used for herbal infusions'. Since the usage of dismissed terms is not allowed, they were grouped only in the Master hierarchy under the term A04KP 'Groups only in the Master hierarchy because dismissed-not in use' and their applicability was changed into optional. This leads to avoid their use but at the same time it enables their tracing.

Table 6: Terms that were dismissed in the tea and herbal infusion sections

Code	Name
A0C6C ^(a)	Tea ingredients (RPC derivatives)
A0ETA ^(a)	Flowers used for infusions
A0ETH ^(a)	Roots used for infusions

(a): See section 3.3 for further details on amendments on tea and herbal infusion sections

In the case of other nine terms, the applicability was changed into optional in the domains in which they were present, thus making them candidates for future deprecation, though not dismissing them in this maintenance. The maintenance technical working group took this preliminary decision since some terms (shown in Table 7) were inconsistent with the logic of the system (mostly as a legacy of previous classifications) while some other (shown in Table 8) identify foods apparently not at all –or very rarely- marketed as such. In the future, after checking the actual use in the existing EFSA databases, deleting these terms will be considered.

As an example of inconsistency with the logic of FoodEx2, in the context of the FoodEx2 classification system some processes are considered not changing the nature of foods from raw commodity to derivative/ingredient; they cannot be used as implicit facets for creating a new derivative FoodEx2 group (for more details see section 3.1.2 of 'The food classification and description system FoodEx2 revision 2' (EFSA, 2015)). Following this logic, for example the process 'boiling' does not create a new nature, thus it can be added to a base term as facet descriptor, but should not be used to create a new food group. Consequently, 'boiled potato' (see code A011P in Table 7) should not exist as base term and should be coded by the users as **A00ZT#F28.A07GL** (i.e. Potatoes, PROCESS=Boiling) i.e. applying the process to the raw commodity.

As an example of foods for which it is unsure if they really exist, some terms included in the section 'Herbal and other non-tea infusions' are apparently not marketed as single herbal/fruit infusion ingredients. Therefore, no significant consumption can be assumed. For example, Jasmine flowers are not used as single ingredient of an infusion; they are mainly used as flavouring ingredient of tea (*Camellia sinensis*), and this tea is marketed as 'Jasmine tea', thus the term A03LJ 'Jasmine infusion', identifying a food item actually not commonly (or not at all) consumed, could lead to an ambiguity for the coder. In order to be consistent with the logic behind the system and in order to avoid misuse of terms, the codes included in Tables 7 and 8 have been marked as optional and their use is discouraged.

Table 7: Terms marked as optional. Instead of them it is rather suggested to use the raw commodity as base term and specify the process facet

Code	Name
A011P	Potato boiled
A011R	Potato baked
A003E	Rice grain, parboiled

Table 8: Terms marked as optional because apparently not existing on the market as such; further investigation will clarify whether to deprecate or reintegrate them.

Code	Name
A03LJ	Jasmine infusion
A03LM	Rose infusion
A03LQ	Strawberry leaves infusion
A03LV	Ginkgo infusion
A03LX	Valerian root infusion
A03LY	Ginseng root infusion

3.5. Revision of applicability or position of some terms in specific hierarchies and/or facets

During the 2015 annual maintenance, refinements on the applicability (i.e. the usage) of FoodEx2 terms and facet descriptors in specific hierarchy/facet domains were implemented. As described in chapter 3.2 of the report on FoodEx2 revision 2 (EFSA, 2015), the food list is organized in 8 hierarchies which are relevant in different food safety domains and the facet descriptors are divided in 28 facets. Most FoodEx2 codes are present in more than one hierarchy and/or facet because they can be relevant in different domains and/or they can be used as facet descriptors.

The term A06TK 'Smoke flavour', already present in the Master, Reporting, Zoonoses and Exposure hierarchies and in the Ingredient facet, was also added in the Process facet as child of A07JV 'Smoking' since some smoked products may be obtained with smoke flavour.

The term A000J 'Grains and grain-based products', a term used in the Exposure hierarchy, was also included in the Reporting hierarchy and in the Ingredient facet in order to group together A0EZF 'Cereal grains and similar and primary derivatives thereof' and A0EZV 'Cereal dough-based products'.

The term A07VP 'Ready-to-eat', used so far for microbiological purposes only in the Intended-use facet was also added in the Qualitative-info facet in order to enable the use of this descriptor in a broader sense in domains such as food consumption where it is needed to identify prepared or partially prepared composite foods without the intention of providing any microbiological information.

Some live plants present so far only as food source were also added in the Botanicals hierarchy, since they were found to be relevant in this domain as well.

In some cases, editorial changes were applied, like in the case of facet descriptors which had been added in the Facets section of the Master terminology but whose addition in the relevant facet hierarchy had been forgotten. These editorials are not detailed in the present report but they are reflected in the database and documented in the Appendix 'FoodEx2_may2016.xlsx' containing the complete content of the FoodEx2 system.

While refining the FoodEx2 content and following the logic behind the relationship between the terms in the hierarchical tree of the system (parent-child structure), some improvements were made in the position of terms within specific hierarchies. In particular, the position in the hierarchy tree was changed since a better parent-child relationship was found.

The main amendments regarding the position of terms within the structure of the system are the following:

- The term A04NS 'Other desserts spoonable', former child of A02PV 'Dairy desserts spoonable', was moved under the more generic term A0C68 'Spoonable desserts and ice creams (generic)', since it includes any type of other desserts spoonable, neither water nor dairy based;
- The terms A00EC 'Wheat germs rolled flakes' and A00ED 'Wheat bran rolled flakes', previously children of A04LJ 'Cereal rolled grains', were grouped under A00EF 'Processed wheat-based flakes' since they identify breakfast cereals made by flaking wheat grains with a wet process similar to corn flakes;
- The term A0F1J 'Edible casings', which describes any type of edible casing and wrapper (both artificial and natural) was moved under A048P 'Miscellaneous agents for food processing', in the Exposure hierarchy; so far this term was grouped under A01QR 'Meat and meat products', which

might suggest that it describes only edible casings made from animal intestine, but this is not always the case;

- The term A03YA 'Veggie pot pie', former child of A03XX 'Vegetable based dishes', was moved under A0CEN 'Savoury pies and tarts' since they are composite food items of the same nature;
- The two terms A024J 'Fresh spiced sausages in casing' and A024K 'Fresh bulk sausages' were moved from A024H 'Italian-style sausage' to A024G 'Fresh raw sausages', since they do not describe only sausages typically made in Italy;
- The terms A0F2E 'Coffee beverage-preparation, powder' and A03HH 'Cocoa beverage-preparation, powder' were grouped together in the Exposure hierarchy under A0C6D 'Hot drinks and infusions composite ingredients';
- The term A00XQ 'Lemon myrtle (fresh herb)' was moved under A04MA 'Aromatic herbs' to distinguish it from the term A0D4K 'Lemon myrtle infusion leaves';
- It was decided to group the term A0FAQ 'Japanese seabass', describing fishes of the taxonomic group *Lateolabrax japonicas* (Percichthyidae family), under the generic term A029S 'Miscellaneous coastal marine fishes' since the former parent term, i.e. A029T 'Sea bass', describes fishes belonging to a different taxonomic family (i.e. Serranidae family);
- The term A027A 'Bluegill sunfish, or bluegill bream' was moved under A04RN 'Miscellaneous freshwater fishes' according to the International Standard Statistical Classification for Aquatic Animals and Plants (ISSCAAP)⁴ used to organize the section 'fish meat' in the FoodEx2 revision2 (EFSA, 2015);
- The term A00LB 'Lollo rosso' was moved under A00LA 'Cutting lettuces', which includes raw commodities of *Lactuca sativa* L. var. *crispa*, since it describes the red leaf variants of the same species.

3.6. Additional activities on the FoodEx2 database

In addition to the annual maintenance, three on-going specific activities have an impact on the content of the FoodEx2 terminology:

1. Mapping FoodEx2 → FoodEx1;
2. Revision of the implicit facets;
3. Revision and enrichment of the scope notes.

The development of a possibly complete bi-directional mapping between FoodEx2 and FoodEx1 is on-going and near to completion. This exercise was performed in the past and the FoodEx1 codes corresponding to each FoodEx2 code were embedded, where available, in the FoodEx2 terminology database, in the FoodExOldCode field. Since the evolution of FoodEx2 between revision 1 and revision 2 was relevant, the codes mapping to FoodEx1 needed a full update. The activity is almost completed, therefore most of the revised code were recently added or updated in FoodEx2. These changes are present in the database together with those deriving from the annual maintenance and will be described in a separate report.

The implicit facets are being revised and completed. This activity is still on-going, but the changes and additions done so far are already included in the FoodEx2 database as in the case of the mapping mentioned in the previous paragraph. Once this activity will be completed, it will be documented in a separate report.

An activity that is planned but has not yet started is the complete revision of the scope notes with the purpose of improving the information they provide and eventually also including external links. This activity will impact on the corresponding field ('Scopenotes') of FoodEx2.

⁴ Food and Agriculture Organization of the United Nations (FAO), Fisheries and Aquaculture Department, ISSCAAP categories. Available online: ftp://ftp.fao.org/FI/STAT/DATA/ASFIS_structure.pdf; accessed on 8 January 2016.

4. Conclusions and recommendations

In 2011 the first revision of the food classification and description system FoodEx2 developed by EFSA was released. During the following four years the system was intensively tested and, based on the outcomes of the texting phase, in 2015 the second revision of FoodEx2 was published by EFSA. The second revision achieved a good level of comprehensiveness and stability; however, it was recommended to carry out an annual maintenance cycle in order to meet new needs required in different food safety domains. Continuous changes in the food market and the experience gained by users and stakeholders imply the evolution of the system.

Based on suggestions provided by users and stakeholders and requests for addition of missing terms, the 2015 FoodEx2 annual maintenance was carried out:

- New terms (271) were added in different hierarchies and facets;
- The names of some existing terms were changed to better specify the scope of the term or to expand its meaning;
- Alias – included terms were added to existing terms to clarify their intended use;
- A major revision was undertaken in the areas of tea and herbal infusion materials and beverages;
- Three double terms were dismissed;
- The use of nine terms was changed into optional and the possibility of deprecating them will be investigated;
- The applicability of some terms was extended to further hierarchies and/or facets;
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found.

While carrying out the 2015 annual maintenance, some improvements on the scope of the terms (i.e. the description of the food group recorded in the field 'Scopenotes') and some amendments on the implicit facets were also carried out. Furthermore, the mapping between FoodEx1, which is the first food classification and description system developed by EFSA intended to be dismissed in the future, and FoodEx2 terms was performed. The complete mapping list has been integrated in FoodEx2 in the Correlated Codes section, in the field 'FoodexOldCode', and is available in the FoodEx2 browser tool. However, these further activities related to FoodEx2 do not fall under the scope of the annual maintenance, thus they are not detailed in the present report. These projects are currently on going and they will be documented and published in the near future.

The present report documents the yearly update of FoodEx2; therefore it should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described. These are essential elements to perform a correct coding with a reasonable level of detail.

The complete content of the FoodEx2 system is available as an Appendix to this document both as exported spreadsheet ('FoodEx2_may2016.xlsx') and as catalogue file ('FoodEx2_may2016.catalog').

The experience gained in this year confirms the need of an annual maintenance and updating cycle for FoodEx2 in order to take into account the evolution of the food market and the new needs in classifying food in the different food safety domains.

References

- EFSA (European Food Safety Authority), 2011a. Report on the development of a food classification and description system for exposure assessment and guidance on its implementation and use. EFSA Journal 2011;9(12):2489, 84 pp., doi:10.2903/j.efsa.2011.2489
- EFSA (European Food Safety Authority), 2011b. The food classification and description system FoodEx 2 (draft revision 1). Supporting Publications 2011:215, 438 pp. Available online: <http://www.efsa.europa.eu/en/supporting/doc/215e.pdf>
- EFSA (European Food Safety Authority), 2015. The food classification and description system FoodEx2 (revision 2). Supporting publication 2015:EN-804. 90 pp.
- THIE (Tea & Herbal Infusions Europe), 2014. Compendium of Guidelines for Tea (*Camellia sinensis*). (3) 43 pp.

Glossary and Abbreviations

Base term	The term describing a food group in a FoodEx2 hierarchy and used as the starting point and principal descriptor in the FoodEx2 code.
Botanicals hierarchy	Hierarchy included in FoodEx2 for reporting data related to plants relevant to the EFSA's compendium of botanicals.
Corex	The field of the FoodEx2 catalogue that records the role of each food group in data reporting.
DATA Unit	Evidence Management Unit of EFSA.
Domain	Domains represent different focus subjects inside the food safety system. Each domain is a specific view of the food chain restricted by e.g. regulations or other commonly accepted ideas. Examples of domains are pesticides, zoonoses, contaminants, additives, nutrients. Different domains need different hierarchies in order to handle classification issues.
EFSA	European Food Safety Authority.
Exposure hierarchy	Food hierarchy included in FoodEx2 arbitrarily defined based on experiences in exposure assessment. It particularly focuses on the need of grouping food items for exposure calculations and it includes only the food section. It is also the preferred hierarchy for reporting consumption data.
Facets	Collections of descriptors used to add further detail to the information provided by the base term, describing properties and aspects of foods from various points of view (e.g. production method, source and ingredient).
Feed hierarchy	Feed hierarchy included in FoodEx2 for reporting data on feed materials.
FoodEx2	Multi-purpose food classification and description system developed by EFSA. Revision 1 of FoodEx2 was published in 2011, while revision 2 was released in 2015. Revision 2 of FoodEx2 replaces the revision 1.
Implicit facet	A facet descriptor that obviously applies to a specific base term and is already pre-recorded in the catalogue.
Ingredient facet	Collections of descriptors defining characterising ingredient(s) in a composite food item.
Master hierarchy	Hierarchy that contains the entire terminology of FoodEx2 which is intended to be used for technical purposes and not for coding.
Part-nature facet	Collections of descriptors defining the generic type of food, i.e. its nature.
Process facet	Collections of descriptors defining the production methods/treatments.
Purpose-of-raising facet	Collections of descriptors defining the reason for keeping, breeding or farming animals. It is relevant in the animal domain.
Qualitative-info facet	Collections of descriptors defining specific characteristic of a food item.
Raw primary commodity (briefly 'raw commodity')	Pieces physically separated from a live source after harvesting (plants) or slaughtering (animals); the separation may also be less invasive, such as picking fruit or collecting milk or eggs. No process changing the nature has been applied. Processes assumed, in the context of this classification, not to change the nature of food (e.g. freezing) can be applied.

Reporting hierarchy	Hierarchy included in FoodEx2 focused on data reporting in a multi-domain context aiming at facilitate the choice of the code and identify the minimum level of detail. It includes sections for food, feed and non-food materials.
RPC	Raw primary commodity.
RPC derivatives/ingredients	Food products obtained from raw commodities (or chemical/microbiological sources) to which processes changing the 'nature' of food have been applied.
Scopenote	The field of the FoodEx2 catalogue that records the description of the term are covered by the code and which are not.
Source facet	Collections of descriptors defining the natural source of a raw commodity, including e.g. live animals and live plants.
Source-commodities facet	Collections of descriptors defining the origin of derivatives of raw commodities.
Statef	The field of the FoodEx2 catalogue that records the position in the food chain and the type of complexity of each food group (state of food).
Surrounding-medium facet	Collections of descriptors defining a surrounding medium. It is mainly relevant for food in packages such as cans, jars or plastic containers with the presence of a surrounding medium.
THIE	Association Tea & Herbal Infusions Europe
Zoonoses hierarchy	Hierarchy included in FoodEx2 focused on reporting biological monitoring (zoonoses) data.

Appendix A – Reference to support files

The two following files containing the updated version of FoodEx2 are provided in addition to the present report:

FoodEx2_may2016.xlsx

FoodEx2_may2016.catalog